AP Computer Science A
Ticket Out the Door
Set 4: Mixed Data Types

Name _____ Period _____

Skill 4.1 Exercise 1	
Treat each box independently. Write the output, or identify the compile-time error and correct it.	
`int cnt = 27.2;` `System.out.println(cnt);`	
`int cnt = (int)27.9;` `System.out.println(cnt);`	
`double d = 5;` `System.out.println(d);`	

`int cnt = 101;`
`double d = cnt;`
`System.out.println(d);`	

`double d = -17.9;`
`int cnt = (int)d;`
`System.out.println(cnt);`	

Skill 4.2 Exercise 1	
Treat each box independently. Write the output, or identify the compile-time error and correct it.	

`int x = 10;`
`int y = 7;`
`System.out.println(y/x);`	

`int x = 10;`
`int y = 7;`
`System.out.println(x/y);`	

`double x = 10;`
`int y = 7;`
`System.out.println(y/x);`	

`int x = 10;`
`int y = 3;`
`double z = x/y;`
`System.out.println(z);`	

`int x = 10;`
`int y = 3;`
`System.out.println((double)x/y);`	

`int x = 10;`
`int y = 3;`
`System.out.println((double)(x/y));`	

`double x = 10.5;`
`int y = 3;`
`System.out.println(x * y);`	

`double z = 10.0 / 3;`
`int i = z;`
`System.out.println(i);`	

`int p = 3;`
`double d = 10.3;`
`int j = (int)5.9;`
`System.out.println(p + p * (int)d - 3 * j);`	

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Skill 4.3 Exercise 1	
Treat each box independently. Write the exact output or name the runtime exception. For an exception, suggest a change that allows the division to run.	

`int x = 0;`
`int y = 0;`
`System.out.println(x/y);`	

`double x = 0;`
`double y = 0;`
`System.out.println(x/y);`	

`int x = 100;`
`double y = 0;`
`System.out.println(x/y);`	

`int x = -100;`
`double y = 0;`
`System.out.println(x/y);`	

Skill 4.4 Exercise 1
Write one Java statement declaring a double constant named SALES_TAX_RATE with the value 0.06.
Is each code segment legal? If illegal, identify the line that causes the error and explain why.

A.
`final double PI = 3.14;`
`int r = 5;`
`double c = 2 * PI * r;`
`PI = c;`
 	
B.
`final int MAX_STUDENTS;`
`MAX_STUDENTS = 30;`
 	
Explanation: _____
_____

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Name _____ Period _____

Skill 4.4 Exercise 2

A class has a snack budget, a number of students, and a price per snack. Assume budget and snackPrice are positive and students is greater than 0. Use these variables in your calculations; do not hard-code the sample results.

```
int budget = 25;
int students = 8;
double snackPrice = 1.75;
```

A. Declare a double constant named TAX_RATE with the value 0.06. Then declare and initialize a double variable priceWithTax that represents the price of one snack including tax.

B. Declare and initialize a double variable budgetPerStudent that represents the budget divided equally among the students. Preserve the fractional part using a cast. Then print budgetPerStudent.

C. Declare and initialize an int variable wholeSnacks that represents the number of whole snacks the class can buy at priceWithTax. Use a cast to discard the fractional part. Then print wholeSnacks.

D. A student writes the following statement for part (b).

```
double budgetPerStudent = (double)(budget / students);
```

What value is stored? Explain why it does not preserve the fractional part, and write a corrected statement.

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